

Core-X: Computer Network

Course Outcomes:

- To understand data communication and network concepts.
- To learn about different communication standards
- To understand different network protocols

Learning Outcomes:

Upon completion of this course, students will be able to:

- Understand concepts on data communication and the use of communication devices
- Learn about analog and digital signals and basic components of data communication
- Learn about errors during data communication & access control mechanisms
- Learn various network protocols and network security issues

Unit I:

Introduction to Data Communications and Network Models: Protocols and Standards, Layers in OSI Models, Analog and Digital Signals, Network Topology, Transmission Modes, Transmission Impairment, Data Rate Limits, Performance, Digital Transmission, Network Devices & Drivers: Router, Modem, Repeater, Hub, Switch, Bridge (fundamental concepts only).

Unit II:

Signal Conversion: Digital-to-Digital Conversion, Analog-to-Digital Conversion, Digital-to-analog Conversion, Analog-to-analog Conversion. Switching Techniques: Packet Switching, Circuit Switching, Datagram Networks, Virtual-Circuit Networks, and Structure of a Switch.

Unit III:

Error Detection and Correction: Parity Check, Checksum, CRC, Error correction technique (Hamming code), Data Link Control: Framing, Flow and Error Control, Noiseless Channels, Noisy channels, (Stop and Wait ARQ, Sliding Window Protocol, Go Back N, Selective Repeat) Point-to-Point Protocol. Access Control: TDM, CSMA/CD, and Channelization (FDMA, TDMA, and CDMA).

Unit IV:

Network Layer: Logical Addressing, IPv4 Addresses, IPv6 Addresses, Subnet, Subnet masking, Virtual-Circuit Networks: Frame Relay and ATM, Transport Layer: Process-Process Delivery: UDP, TCP. Application layers: DNS, SMTP, POP, FTP, HTTP, Basics of WiFi (Fundamental concepts only), and Network Security: Authentication, Basics of Public Key and Private Key Cryptography, Digital Signatures and Certificates (Fundamental concepts only).

Text Book:

- ✓ *Computer Networks, A. S. Tanenbaum, 4th edition, Pearson Education.*

Reference Book:

- ✓ *Data Communications and Networking, Fourth Edition by Behrouza A. Forouzan, TMH.*